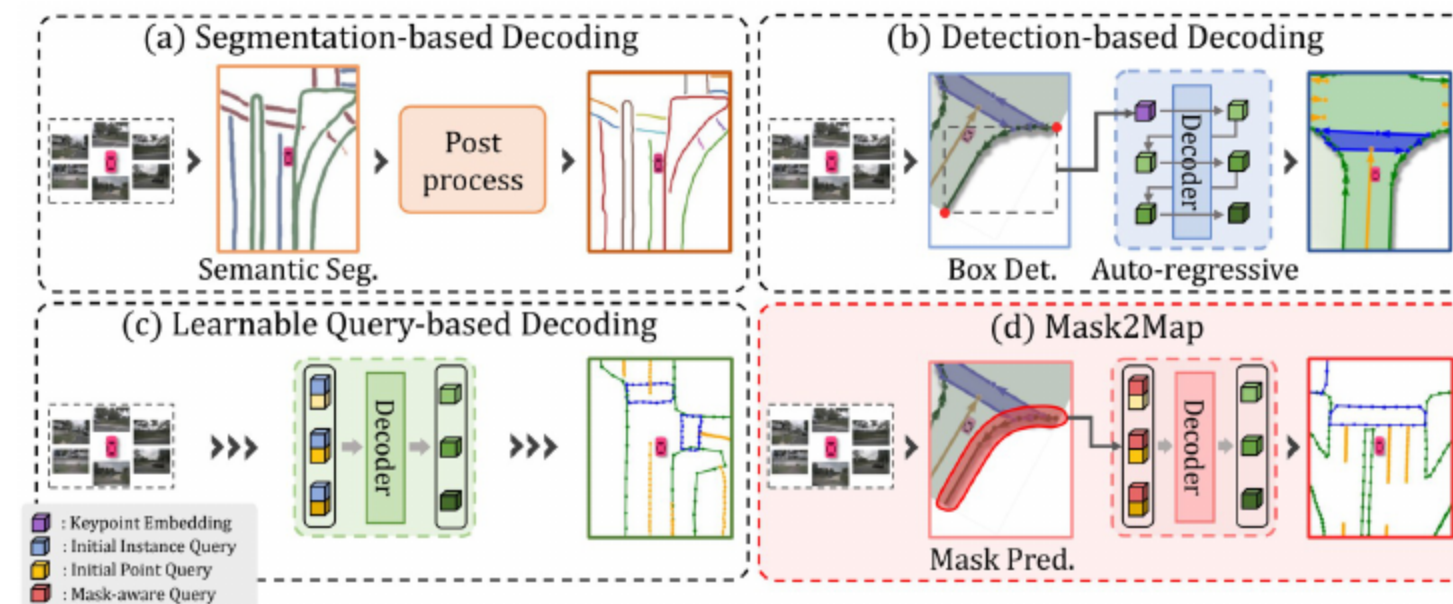


Mask2Map: Vectorized HD Map Construction Using BEV Segmentation Masks

Sehwan Choi, Jungho Kim, Hongjae Shin, Jun Won Choi

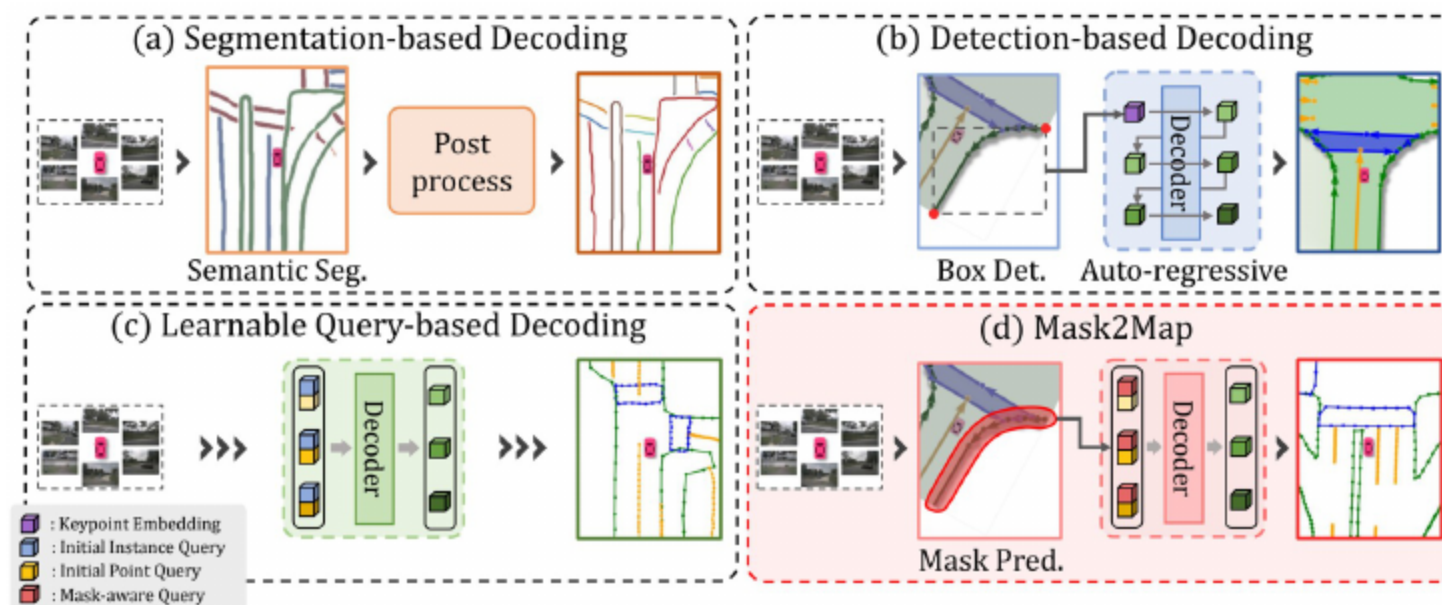
01 Motivation: Shift to online HD maps



- HD maps provide rich semantics and geometry for precise planning and safety [1,8,10,11,19]
- Traditional offline SLAM pipelines are **labor-intensive**, **costly**, and **slow to update** [33,34,42]
- Learning-based online local maps from onboard sensors enable **immediate updates**, **scalable deployment**, and **reduced maintenance**

[Antonello et al. 2017]

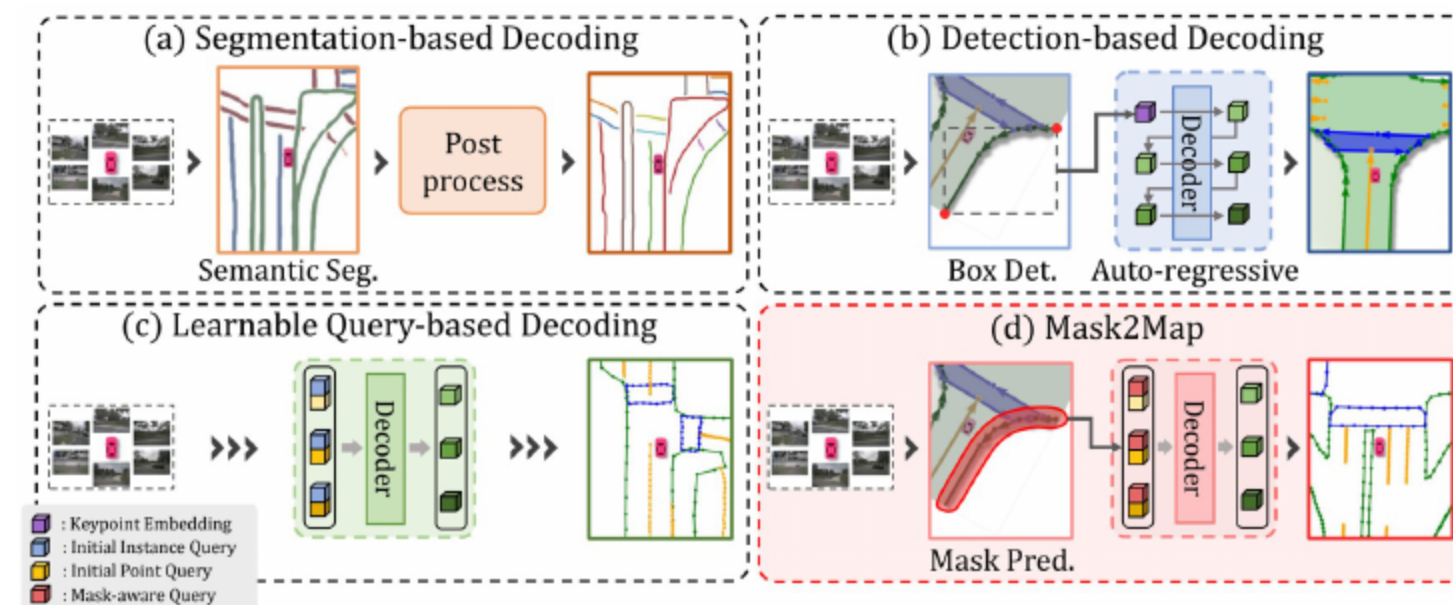
02 Segmentation and query-based decoding



- Segmentation-based: raster outputs lack vector topology; post-processing is **slow and brittle** [17]
- Detection-based: keypoint reliance misses **diverse shapes** and **instance continuity** [24]
- Learnable query-based: scene-agnostic initial queries weaken **semantic grounding** and **geometric fidelity** [9,20,21,38,40]

[Winter 2023]

03 Mask-Aware Queries → vectorized maps

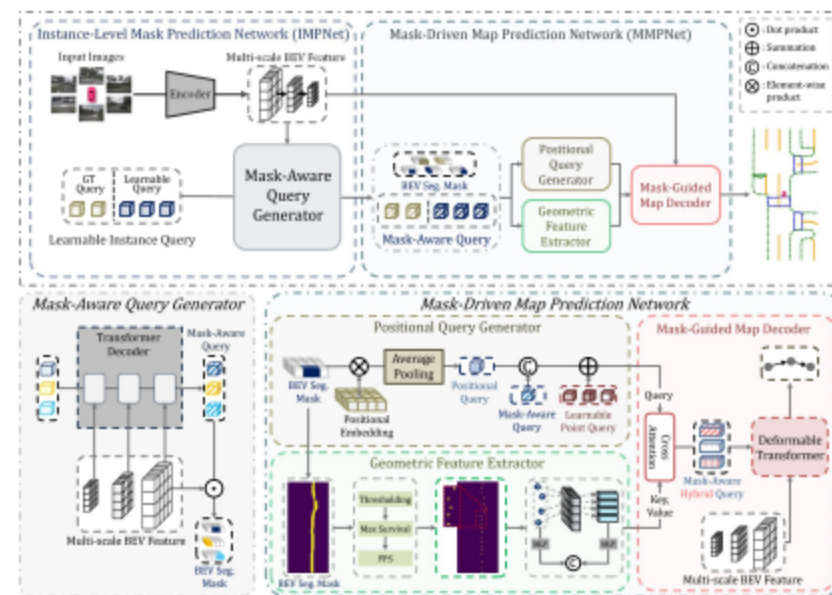


- **IMPNet**: predicts instance-level BEV masks using Mask-Aware Queries adapted from Mask2Former [5]
- **MMPNet**: decodes ordered points with Positional Query Generator and Geometric Feature Extractor
- Global semantics from masks plus local decoder refinement yield **accurate** and **coherent** vectorized HD maps

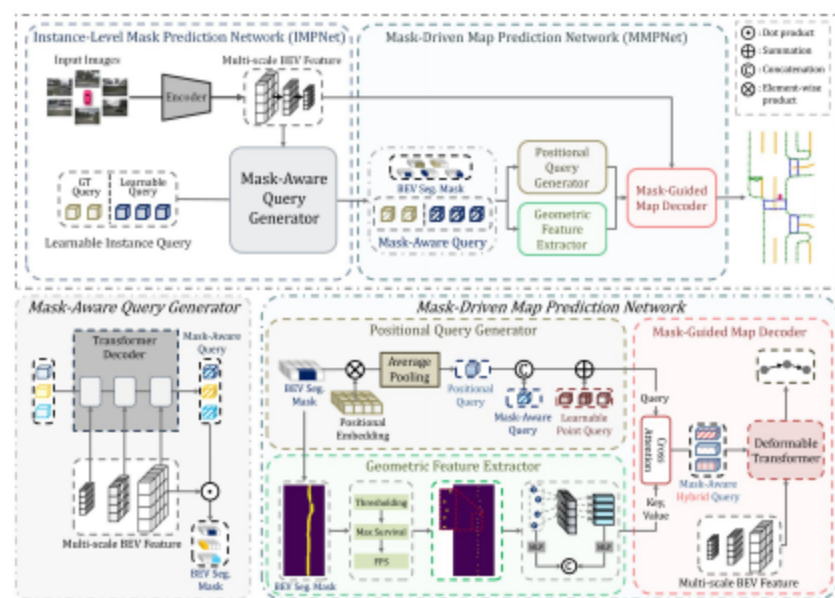
Method	Modality	Backbone	Epoch	AP_{ped}	AP_{diver}	$AP_{boundary}$	mAP	FPS
MapTR [20]	C	R50	24	46.3	51.5	53.1	50.3	15.1
MapVR [40]	C	R50	24	47.7	54.4	51.4	51.2	15.1
PivotNet [9]	C	R50	24	56.2	56.5	60.1	57.6	-
BeMapNet [30]	C	R50	30	57.7	62.3	59.4	59.8	-
MapTRv2 [21]	C	R50	24	59.8	62.4	62.4	61.5	14.1
Ours	C	R50	24	70.6	71.3	72.9	71.6	10.1
VectorMapNet [24]	C	R50	110	36.1	47.3	39.3	40.9	2.2
MapTR [20]	C	R50	110	56.2	59.8	60.1	58.7	15.1
MapVR [40]	C	R50	110	55.0	61.8	59.4	58.8	15.1
MapTRv2 [21]	C	R50	110	68.1	68.3	69.7	68.7	14.1
Ours	C	R50	110	73.6	73.1	77.3	74.6	10.1
VectorMapNet [24]	C+L	R50 & PP	110	37.6	50.5	47.5	45.2	-
MapTR [20]	C+L	R50 & Sec	24	55.9	62.3	69.3	62.5	6.0
MapVR [40]	C+L	R50 & Sec	24	60.4	62.7	67.2	63.5	6.0
MapTRv2 [21]	C+L	R50 & Sec	24	65.6	66.5	74.8	69.0	5.8
Ours	C+L	R50 & Sec	24	76.5	76.6	82.1	78.4	4.1

- nuScenes (camera): 71.6 mAP at 24 epochs, 74.6 at 110; **+10.1** over MapTRv2 at 24 epochs
- Raster metric: 54.7 mAP versus MapTRv2 (**+18.0**) on nuScenes
- Argoverse2 (camera, R50): **+4.1** mAP over MapTRv2
- Contributions: **Mask-Aware Queries in BEV**, **PQG**, **GFE**, and inter-network denoising for matching consistency

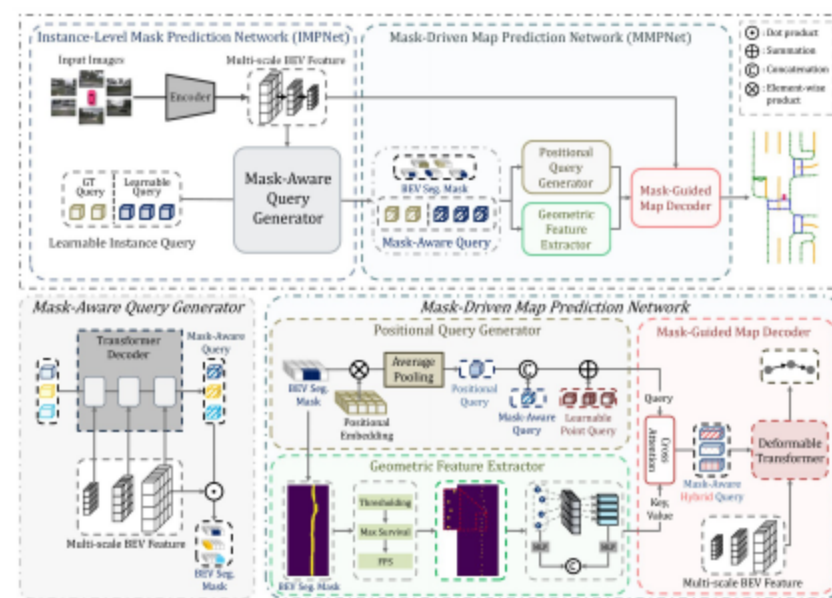
[Kim et al. 2021]



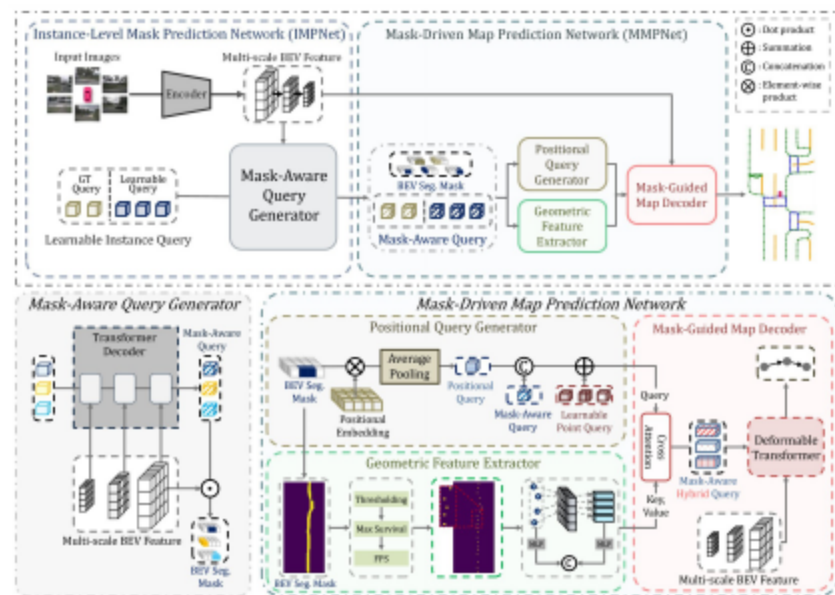
- IMPNet: multi-scale BEV features drive Mask-Aware Queries and instance BEV masks
- MMPNet: PQG adds **positional cues**; GFE adds **geometric cues**; Mask-Guided Decoder fuses them
- Per instance outputs: **class score** and **ordered BEV point sequence** for vectorized maps



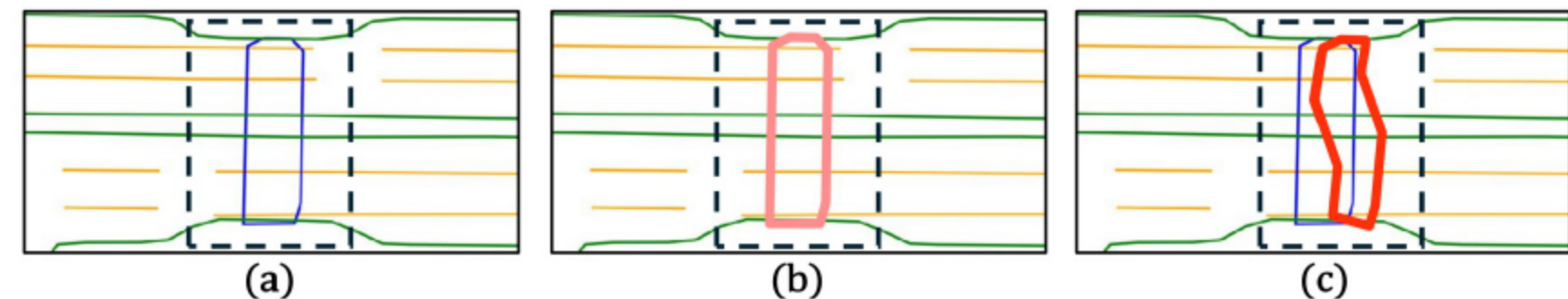
- Inputs: camera via LSS, LiDAR via voxel or SECOND, or camera-LiDAR fusion
- Deformable Transformer encoder aggregates **multi-scale BEV features** efficiently
- Mask Transformer iteratively decodes **instance masks** and updates **queries** for stronger semantics



- Sparsify BEV masks and inject 2D sinusoidal positional embeddings to encode **location**
- Aggregate masked embeddings to instance-level **positional queries** via pooling
- Form point-level queries by combining positional queries with **learnable point queries**



- Sample keypoints from masks using Max Survival windows then farthest point sampling to select pixels
- Pool BEV features at sampled coordinates and encode coordinates with an MLP to get **point-wise geometry**
- Cross-attend PQG queries to GFE features; deformable decoding over **multi-scale BEV** for robust vectors



- Inject noisy ground-truth queries (class-embedded with flip probability) and perturbed ground-truth masks
- Supervise noisy ground-truth queries directly through both networks without bipartite matching
- Improves **consistency** and **convergence**, yielding **higher mAP** and **better utilization**

Method	Modality	Backbone	Epoch	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP	FPS
MapTR [20]	C	R50	24	46.3	51.5	53.1	50.3	15.1
MapVR [40]	C	R50	24	47.7	54.4	51.4	51.2	15.1
PivotNet [9]	C	R50	24	56.2	56.5	60.1	57.6	-
BeMapNet [30]	C	R50	30	57.7	62.3	59.4	59.8	-
MapTRv2 [21]	C	R50	24	59.8	62.4	62.4	61.5	14.1
Ours	C	R50	24	70.6	71.3	72.9	71.6	10.1
VectorMapNet [24]	C	R50	110	36.1	47.3	39.3	40.9	2.2
MapTR [20]	C	R50	110	56.2	59.8	60.1	58.7	15.1
MapVR [40]	C	R50	110	55.0	61.8	59.4	58.8	15.1
MapTRv2 [21]	C	R50	110	68.1	68.3	69.7	68.7	14.1
Ours	C	R50	110	73.6	73.1	77.3	74.6	10.1
VectorMapNet [24]	C+L	R50 & PP	110	37.6	50.5	47.5	45.2	-
MapTR [20]	C+L	R50 & Sec	24	55.9	62.3	69.3	62.5	6.0
MapVR [40]	C+L	R50 & Sec	24	60.4	62.7	67.2	63.5	6.0
MapTRv2 [21]	C+L	R50 & Sec	24	65.6	66.5	74.8	69.0	5.8
Ours	C+L	R50 & Sec	24	76.5	76.6	82.1	78.4	4.1

- Datasets: nuScenes and Argoverse2 with multi-camera and LiDAR, large-scale annotated frames
- Tasks/classes: Pedestrian Crossing, Lane Divider, Road Boundary within a fixed BEV range
- Metrics: Chamfer AP, raster AP, and query utilization for inter-network consistency

Method	Modality	Backbone	Epoch	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP	FPS
MapTR [20]	C	R50	24	46.3	51.5	53.1	50.3	15.1
MapVR [40]	C	R50	24	47.7	54.4	51.4	51.2	15.1
PivotNet [9]	C	R50	24	56.2	56.5	60.1	57.6	-
BeMapNet [30]	C	R50	30	57.7	62.3	59.4	59.8	-
MapTRv2 [21]	C	R50	24	59.8	62.4	62.4	61.5	14.1
Ours	C	R50	24	70.6	71.3	72.9	71.6	10.1
VectorMapNet [24]	C	R50	110	36.1	47.3	39.3	40.9	2.2
MapTR [20]	C	R50	110	56.2	59.8	60.1	58.7	15.1
MapVR [40]	C	R50	110	55.0	61.8	59.4	58.8	15.1
MapTRv2 [21]	C	R50	110	68.1	68.3	69.7	68.7	14.1
Ours	C	R50	110	73.6	73.1	77.3	74.6	10.1
VectorMapNet [24]	C+L	R50 & PP	110	37.6	50.5	47.5	45.2	-
MapTR [20]	C+L	R50 & Sec	24	55.9	62.3	69.3	62.5	6.0
MapVR [40]	C+L	R50 & Sec	24	60.4	62.7	67.2	63.5	6.0
MapTRv2 [21]	C+L	R50 & Sec	24	65.6	66.5	74.8	69.0	5.8
Ours	C+L	R50 & Sec	24	76.5	76.6	82.1	78.4	4.1

- Camera-only: 71.6 mAP at 24 epochs, 74.6 at 110, surpassing MapTRv2 by **+10.1** and **+5.9**
- Camera plus LiDAR fusion: **+9.4** mAP over MapTRv2 with R50 and SECOND
- Raster metric: **+18.0** mAP over MapTRv2 with camera-only inputs

12 State-of-the-art raster mAP†

Method	Epoch	$AP_{ped}^\dagger$	$AP_{divider}^\dagger$	$AP_{boundary}^\dagger$	$mAP^\dagger$
MapTR [20]	24	32.4	23.5	17.1	24.3
MapVR [40]	24	37.5	33.1	23.0	31.2
MapTRv2* [21]	24	49.9	34.7	25.7	36.7
Ours	24	62.9	52.3	48.9	54.7

- Camera-only, R50: **+4.1** mAP over MapTRv2 on the validation set
- Consistent gains across **pedestrian crossings**, **dividers**, and **boundaries**
- Demonstrates **robust generalization** across datasets and scenarios

13 Main components and denoising effects

	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP
Baseline	30.1	41.5	46.6	39.4
+IMPNet	42.7	45.3	48.1	45.3
+MMPNet	45.7	47.0	54.4	49.1
+Denoising	52.9	55.5	58.5	55.6

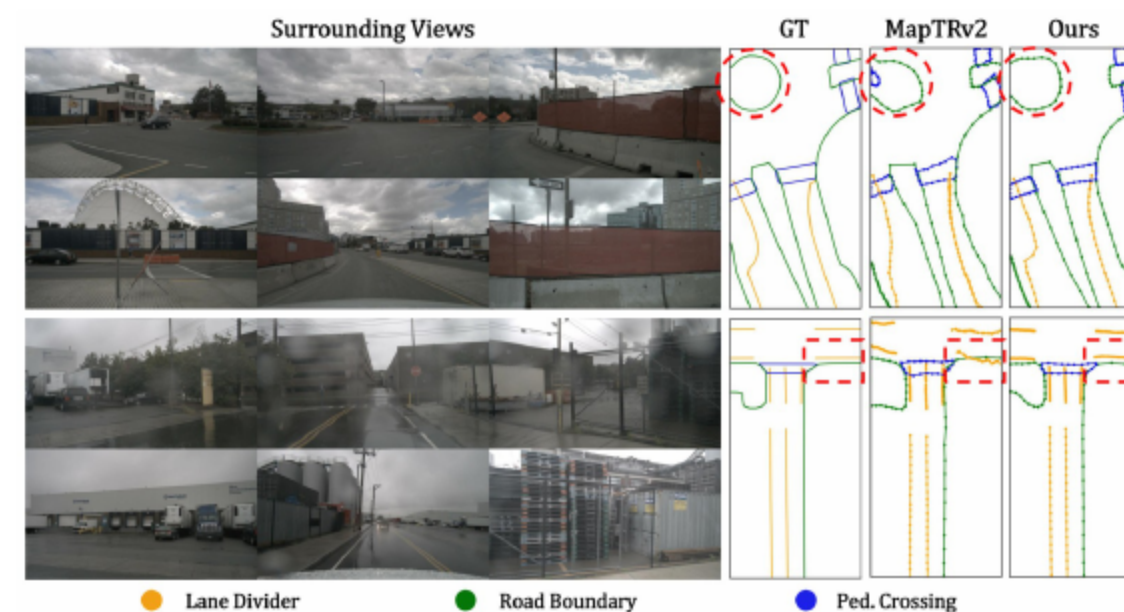
- Adding IMPNet: **+5.9** mAP; adding MMPNet: **+3.8** mAP (cumulative)
- Inter-network denoising: **+6.5** mAP; query utilization improves from 24.7 to **74.7**
- Denoising stabilizes query-to-ground-truth matching across networks and layers

14 PQG and GFE contributions; map noise

PQG	GFE	AP_{ped}	$AP_{divider}$	$AP_{boundary}$	mAP
		47.6	51.1	53.8	50.8
	✓	49.7	54.2	57.8	53.9
✓		49.9	55.6	57.2	54.2
✓	✓	52.9	55.5	58.5	55.6

- PQG: **+3.4** mAP; GFE: **+3.1** mAP; combined: **+4.8** mAP (complementary)
- Map noise on ground-truth masks in denoising yields **+0.8** mAP over clean masks
- Confirms benefits of **positional** and **geometric** cues plus **robust** noisy supervision

15 nuScenes examples vs. MapTRv2



- Cleaner and **more continuous** lane dividers and road boundaries
- Improved **instance consistency** and fewer **artifacts** across scenes
- Better alignment of predicted vectors with **ground-truth** structures

Limitations: no temporal fusion—occlusions may **degrade performance**; lower FPS than MapTRv2 due to richer decoding. Future: temporal fusion via query or BEV memory; model compression and optimization to recover **near real-time** speed.

Mask2Map leverages instance-level BEV masks to guide vectorized decoding. PQG adds **positional cues**; GFE injects **geometric structure**; inter-network denoising aligns **matching** and boosts **stability**. The approach achieves new state of the art on nuScenes and Argoverse2.



THANKS!